

SUBRATA BAG

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[Linkedin](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Early-career Application Software Developer with hands-on experience in software development, database-backed application design, and healthcare technology. Proficient in Java, SQL, Python, and C/C++ with practical exposure to full-stack software engineering using REST APIs, Git-based version control, data analysis, and debugging. Built a healthcare technology platform involving AI-based medical report analysis, data processing, health dashboards, and patient workflows — awarded 3rd Rank and Best Innovation Idea among 50+ teams. Experienced in collaborating with stakeholders, translating requirements into working software, and applying root-cause analysis to resolve application defects.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript

Database / DBMS: SQL, PostgreSQL, MongoDB, Supabase, Database Management Systems

Backend / APIs: Node.js, Express.js, REST APIs, JWT Authentication

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS, Three.js, React Three Fiber

AI / Data Analysis: LLM API Integration, AI-based Data Extraction, Data Processing, Data Visualization

Tools / DevOps: Git, GitHub, Postman, Vercel, VS Code

PROFESSIONAL EXPERIENCE

Software Developer Intern — AZMTH Labs Pvt. Ltd.

Feb 2026 – Jun 2026

- Developed and enhanced interactive web application interfaces using React, Three.js, and React Three Fiber, implementing component-based architecture within an existing production codebase.
- Built and integrated reusable UI components into client-facing web applications, collaborating with team members and stakeholders to translate design requirements into functional, responsive software features.
- Debugged rendering, state-management, and performance issues across React components and Three.js scenes, applying root-cause analysis and systematic debugging to resolve application defects.
- Managed code using Git version control, participating in code reviews, branching workflows, and iterating on feedback from senior developers to maintain software quality.

PROJECTS

MediCon — Healthcare Technology Platform | [Live](#) | [GitHub](#)

React, Node.js, Express, MongoDB, PostgreSQL, REST APIs, AI Integration, Git

- Designed and developed a full-stack healthcare application featuring a web dashboard for health data visualization, QR/ID-based doctor identification, budget-based doctor suggestions, and patient health tracking workflows.
- Integrated AI-based risk detection that processes uploaded medical reports and flags potential health concerns, enabling data-driven analysis and decision support for healthcare users.
- Developed 10+ REST API endpoints backed by MongoDB and PostgreSQL for managing patient records, doctor data, and health tracking information — handling data fetching, processing, and representation based on requirements.
- Built an AI-powered healthcare chatbot using API integration for intelligent assistance, and implemented WhatsApp-based alert notifications and Health Connect Android synchronization for health data.
- Collaborated with a team of developers, coordinating tasks, creating estimates, and participating in technical design discussions to deliver the project under hackathon timelines.

Awards: 3rd Rank among 50+ teams at Tech Expo 2025 | Best Innovation Idea Award — Tech Expo 2025 | Hack-O-NIT — Finalist

AlgoMaster — Coding Practice Platform | [Live](#) | [GitHub](#)

React, Node.js, Express, MongoDB, JWT, REST APIs, Judge0, Git

- Developed a full-stack software application for coding practice with user authentication (JWT), role-based admin access control, and a question management system backed by MongoDB database operations.

- Built 8+ REST APIs using Node.js and Express for authentication, question CRUD operations, user profile management, and coding submission workflows, integrating Judge0 for code execution and evaluation.
- Implemented admin panel features for managing programming problems including creation, editing, categorization, and test-case configuration — supporting the complete question lifecycle and data management.
- Debugged API response handling, database query issues, and authentication edge cases during development, applying systematic debugging and root-cause analysis to identify and resolve data inconsistencies.

ACHIEVEMENTS

- **Tech Expo 2025** — 3rd Rank among 50+ teams (MediCon healthcare project)
- **Tech Expo 2025** — Best Innovation Idea Award (MediCon)
- **Hack-O-NIT** — Finalist

EDUCATION

B.Tech — Computer Science and Engineering | CGPA: 7.36

Calcutta Institute of Technology (MAKAUT) | Expected June 2027

Relevant Coursework: Database Management Systems, Object-Oriented Programming (Java), Data Structures and Algorithms, Operating Systems, Software Engineering